多重网格方法（O(N)算法）

§1. 模型问题

(VP) 求 $u \in H_0^1$ s.t. $a(u, v) = (f, v) \quad \forall v \in H_0^1$

M_k：三角剖分 M_k 是 M_{k-1} 一致加密得到的. $k \geq 1$

$V_k = \{v_k \in H_0^1 : v_k|_K \in P_1 \quad \forall K \in M_k\} \quad V_k \supset V_{k-1}$

(FEM) 求 $u_k \in V_k$ s.t. $a(u_k, v_k) = (f, v_k) = (Q_k f, v_k) \quad \forall v_k \in V_k$

$Q_k : L^2 \longrightarrow V_k \qquad (Q_k w, v_k) = (w, v_k) \qquad \forall v_k \in V_k$

$P_k : H_0^1 \longrightarrow V_k \qquad a(P_k w, v_k) = a(w, v_k)$

$A_k : V_k \longrightarrow V_k \qquad (A_k u_k, v_k) = a(u_k, v_k) \quad \forall u_k, v_k \in V_k$

算子形式：$A_k u_k = f_k := Q_k f$（此定义与变分形式相容）

节点基：$\{\phi_k^1, \dots, \phi_k^{n_k}\} \quad n_k = \dim V_k$

$\forall v_k \in V_k \quad v_k = v_k^1 \phi_k^1 + \dots + v_k^{n_k} \phi_k^{n_k} \qquad v_k^i = v_k(x_k^i)$

定义算符 $\sim, \approx : V_k \longrightarrow \mathbb{R}^{n_k}$

$\tilde{v}_k = (v_k^i)_{n_k \times 1} \qquad \tilde{\tilde{f}}_k = ((f_k, \phi_k^i))_{n_k \times 1}$

$\tilde{A}_k = (a(\phi_k^j, \phi_k^i))_{n_k \times n_k}$

矩阵形式 $\tilde{A}_k \tilde{u}_k = \tilde{\tilde{f}}_k$

性质1：$\widetilde{\widetilde{A_k u_k}} = \tilde{A}_k \tilde{u}_k$

$(A_k u_k, \phi_k^i) = a(u_k, \phi_k^i) = \sum_j a(\phi_k^j, \phi_k^i) u_k^j = (\tilde{A}_k \tilde{u}_k)_i$

$h_k = \max\limits_{K \in M_k} h_K \qquad \|v\|_A = a(v, v)^{\frac{1}{2}}$

§经典迭代法

线性迭代法：给定初值 $\tilde{u}^0$

迭代子 一般是 $\tilde{A}_k^{-1}$ 的近似.　　收敛 $\Leftrightarrow$ $\rho(I-\tilde{R}_k\tilde{A}_k)<1$

$n=0,1,2,\dots$,　$\tilde{u}^{n+1}=\tilde{u}^n+\tilde{R}_k(\tilde{f}_k-\tilde{A}_k\tilde{u}^n)$　$\Rightarrow$　$\tilde{u}^{n+1}-\tilde{u}_k=(I-\tilde{R}_k\tilde{A}_k)(\tilde{u}^n-\tilde{u}_k)$

矩阵形式　$\tilde{u}_k=\tilde{u}_k+\tilde{R}_k(\tilde{\tilde{f}}_k-\tilde{A}_k\tilde{u}_k)$

$$\tilde{R}_k=\begin{cases}\omega I & 0<\omega<\frac{2}{\rho(\tilde{A}_k)} \quad \text{Richardson}\\ (\omega)\tilde{D}^{-1} & 0<\omega<\frac{2}{\rho(\tilde{D}_k^{-1}\tilde{A}_k)} \quad \text{(阻尼)Jacobi}\\ (\tilde{D}+\tilde{L})^{-1} & \text{恒收敛} \quad \text{Gauss-Seidel}\end{cases}$$

对称阵　对角线　下三角

$\tilde{A}_k=\tilde{D}+\tilde{L}+\tilde{L}^T$　　$I-\omega\tilde{A}_k$

分量形式　$Ax=b$

$a_{i1}x_1+\dots+a_{in}x_n=b_i$

Jacobi　$x_i^{n+1}=\frac{1}{a_{ii}}\left(b_i-\sum_{j\neq i}a_{ij}x_j^n\right)$

Gauss-Seidel　$x_i^{n+1}=\frac{1}{a_{ii}}\left(b_i-\sum_{j=1}^{i-1}a_{ij}x_j^{n+1}-\sum_{j=i+1}^{n}a_{ij}x_j^n\right)$

收敛速度均为　$1-ch^2$

算子形式：求 $A_ku_k=f_k$ 的线性迭代法：给定 u^0

$u^{n+1}=u^n+R_k(f_k-A_ku^n)$　　$n=0,1,2,\dots$

$\tilde{u}^{n+1}=\tilde{u}^n+\widetilde{R_k(f_k-A_ku^n)}$

$R_k: V_k\longrightarrow V_k.$

$\widetilde{R_kg}=\tilde{R}_k\tilde{\tilde{g}}$　$\Rightarrow$　$R_kg=\sum_{i=1}^{n_k}\left(\sum_{j=1}^{n_k}(\tilde{R}_k)_{ij}(g,\phi_k^j)\right)\phi_k^i$

引理：
$$R_kg=\begin{cases}\sum_i\omega(g,\phi_k^i)\phi_k^i & \text{Richardson.}\\ \omega\sum_{i=1}^{n_k}P_k^iA_k^{-1}g & \text{阻尼 Jacobi}\\ (I-E_k)A_k^{-1}g\end{cases}$$

Jacobi

$P_k^i: H_0^1\longrightarrow \operatorname{span}(\phi_k^i)$

$a(P_k^iv,\phi_k^i)=a(v,\phi_k^i)$　$\Rightarrow$　$\alpha=\frac{a(v,\phi_k^i)}{a(\phi_k^i,\phi_k^i)}$

$\alpha\phi_k^i$

$P_k^iv=\frac{a(v,\phi_k^i)}{a(\phi_k^i,\phi_k^i)}\phi_k^i$　　$(A_kv_k,w_k)=a(v_k,w_k)$

$R_kg=\sum_i\omega\frac{(g,\phi_k^i)}{a(\phi_k^i,\phi_k^i)}\phi_k^i=\sum_i\omega P_k^iA_k^{-1}g$　$(g,\phi_k^i)=(A_kA_k^{-1}g,\phi_k^i)=a(A_k^{-1}g,\phi_k^i)$

$v=A_k^{-1}g$

Gauss - Seidel

Gauss-S 分块 红黑树？

块迭代

$E_k = (I - P_k^{n_k}) \cdots\cdots (I - P_k^1)$

$I - R_k A_k = \begin{cases} I - \omega \sum P_k^i & \text{阻尼 Jacobi} \\ E_k & \text{G-S（施瓦兹迭代）} \end{cases}$

收敛速度慢 但具有磨光性质

例：

磨光性质：$\begin{cases} -u'' = f \\ u(0) = u(1) = 0 \end{cases}$

等距剖分

N 个　　$x_i = ih$，$h = \frac{1}{N}$

自由度：$N-1$

中心差商

线性有限元方法：$\frac{-u_{i-1} + 2u_i - u_{i+1}}{h^2} = f_i$　$\tilde{A}\tilde{u} = \tilde{\tilde{f}}$

$\begin{cases} -u'' = \lambda u \\ u(0) = u(1) = 0 \end{cases}$

$u_k = \sin k\pi x$

$\lambda_k = k^2\pi^2$

$\tilde{A} = \begin{pmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{pmatrix}$　　$\hat{\phi}_k = (\sin k\pi x_i)_{N-1 \times 1}$

$Au = \lambda u$.

$\frac{-u_{i-1} + 2u_i - u_{i+1}}{h^2} = \lambda_{k,h} u_i$

$\Rightarrow -\sin k\pi x_{i-1} + 2\sin k\pi x_i - \sin k\pi x_{i+1} = 2\sin k\pi x_i - 2\sin k\pi x_i \cos k\pi h$.

$\lambda_{k,h} = \frac{2 - 2\cos k\pi h}{h^2} = \frac{4\sin^2\frac{k\pi h}{2}}{h^2}$

Richardson 迭代：$\tilde{u}^{n+1} = \tilde{u}^n + \frac{1}{\rho(\tilde{A})}(\tilde{\tilde{f}} - \tilde{A}\tilde{u}^n)$

$\tilde{u} = \tilde{u}$

$\tilde{u}^{n+1} - \tilde{u} = (I - \frac{1}{\rho(\tilde{A})}\tilde{A})(\tilde{u}^n - \tilde{u})$

$\tilde{u}^n - \tilde{u} = (I - \frac{1}{\rho(\tilde{A})}\tilde{A})^n(\tilde{u}^0 - \tilde{u})$

设 $\tilde{u}^0 - \tilde{u} = \sum_{i=1}^{n-1} \alpha_i \tilde{\phi}_i$

$\tilde{u}^n - \tilde{u} = \sum_{i=1}^{n-1} \alpha_i (1 - \frac{\lambda_{i,h}}{\lambda_{n-1,h}})^n \tilde{\phi}_i$

引理：阻尼 Jacobi，$\omega > 0$ 足够小 $\exists\ \alpha_0 > 0$ s.t.　　G-S 也对

$(I - P_{k-1})K_k^m v \leq \frac{\alpha_0}{m}(\|v\|_A^2 - \|K_k^m v\|_A^2)$

前一层网格投影

其中 $K_k = I - R_k A_k$.

§3 多重网格 V-cycle 算法

思想：在当前网格磨光在粗网格上求解 校正

$A_k u_k = f_k$.

MG: 给定 $u^0 \in V_k$ $\quad u^{n+1} = u^n + B_k(f_k - A_k u^n)$ $\quad n = 0, 1, 2, \ldots$

$u_k = u_k + B_k(f_k - A_k u_k)$

MGP: 定义 B_k：给定 m, R_k：设 $B_1 = A_1^{-1}$

$B_{k-1}: V_{k-1} \longrightarrow V_{k-1}$ 已定义 $\quad g \in V_k$ 如下定义：$B_k g$

$A_k y = g$ $\quad$ 0为初值 $\Leftrightarrow B_k g$.

(i) 前磨光：$y^0 = 0$ $\quad y^{i+1} = y^i + R_k(g - A_k y^i)$ $\quad i = 0 \ldots m-1$

y^m $\quad$ Res $g - A_k y^m$

$A_k e = g - A_k y^m$

求 $e \in V_k$. $a(e, v_k) = (g - A_k y^m, v_k)$

求 $e \in V_{k-1}$. $a(e, v_{k-1}) = (g - A_k y^m, v_{k-1})$

$A_{k-1} e = Q_{k-1}(g - A_k y^m)$

(ii) 粗网格校正：$\quad e = B_{k-1} Q_{k-1}(g - A_k y^m)$ $\quad y^{m+1} = y^m + e$

(iii) 后磨光：$\quad y^{i+1} = y^i + R_k^t(g - A_k y^i)$ $\quad i = m+1, \ldots, 2m$

$(R_k v_k, w_k) = (v_k, R_k^t w_k)$

令 $\underline{B_k g = y^{2m+1}}$ $\quad$ 为何是定义 $B_k g$ $\quad$ 若效果好 则 $B_k \sim A_k^{-1}$

A_k 条件数差 则 CG 速度慢

此时预条件于 $B_k A_k y = B_k g$ $\quad$ PCG

$u^{n+1} - u_k = \underbrace{(I - B_k A_k)}_{\text{误差传播算子}}(u^n - u_k)$

$y - y^m = (\underbrace{I - R_k A_k}_{K_k})^m (y - y^0)$ $\quad$ 可能较大

$= K_k^m (y - y^0)$ $\quad K_k = I - R_k A_k$

$e = B_{k-1} Q_{k-1} A_k (y - y^m)$, $\quad y - y^{m+1} = (I - B_{k-1} Q_{k-1} A_k)(y - y^m)$

可以考虑对称 G-S

$y - y^{2m+1} = (I - R_k^t A_k)^m (y - y^{m+1})$ $\quad K_k^* = I - R_k^t A_k$

$(I - B_k A_k) y = y - B_k g = y - y^{2m+1} = (K_k^*)^m (I - B_{k-1} \underline{Q_{k-1} A_k}) K_k^m y$

$(Q_{k-1} A_k v_k, w_{k-1}) = (A_k v_k, w_{k-1}) = a(v_k, w_{k-1}) = a(P_{k-1} v_k, w_{k-1}) = (A_{k-1} P_{k-1} v_k, w_{k-1})$

$Q_{k-1} A_k = A_{k-1} P_{k-1}$ $\quad$ on V_k

$I - B_kA_k = (K_k^*)^m (I - B_{k-1}A_{k-1}P_{k-1}) K_k^m$

引理: $I - B_kA_k = (K_k^*)^m (I - P_{k-1} + (I - B_{k-1}A_{k-1})P_{k-1}) K_k^m$

K_k 与 K_k^* 关系

$\forall v, w \in V_k$

$a(K_k v, w) = (A_k(I - R_kA_k)v, w) = (I - R_kA_k v, A_k w) = (A_k v, w) - (A_k v, R_k^t A_k w)$

$= (A_k v, (I - R_k^t A_k) w)$

$= a(v, K_k^* w)$

引理: 阻尼 Jacobi 法 $R_k^t = R_k$. $K_k^* = K_k$

且 $(I - B_kA_k)^* = I - B_kA_k$. $B_k^t = B_k$

$R_k = w \sum_i P_k^i A_k^{-1}$

局部 H^1 投影.

$a(P_k^i v, \phi_k^i) = a(v, \phi_k^i)$

$P_k^i : H_0^1 \longrightarrow span(\phi_k^i)$

$(R_k v, w) = w \sum_i (P_k^i A_k^{-1} v, w)$

$= w \sum_i (A_k P_k^i A_k^{-1} v, A_k^{-1} w)$

$= w \sum_i a(P_k^i A_k^{-1} v, P_k^i A_k^{-1} w)$ 利用局部 H^1 投影定义

$a((I - B_kA_k)v, w) = a((K_k^*)^m (I - P_{k-1} + (I - B_{k-1}A_{k-1})P_{k-1}) K_k^m v, w)$

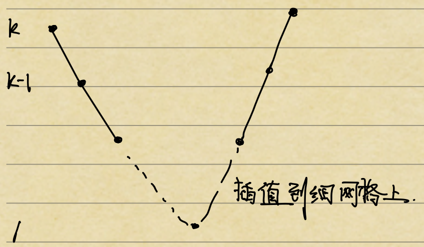
$= a(((I - P_{k-1}) + (I - B_{k-1}A_{k-1})P_{k-1}) K_k^m v, K_k^m w)$

$= a((I - P_{k-1}) K_k^m v, K_k^m w) + a((I - B_{k-1}A_{k-1})P_{k-1} K_k^m v, K_k^m w)$

Galerkin 正交

$= a((I - P_{k-1}) K_k^m v, (I - P_{k-1}) K_k^m w) + a((I - B_{k-1}A_{k-1})P_{k-1} K_k^m, P_{k-1} K_k^m w)$

$B_1 = A_1^{-1}$ ⇒ 初始对称.



可以写成非递归形式效率更高.?

定理：（阻尼 Jacobi 磨光的MG） $\omega>0$ 足够小，则 $\exists\,\alpha>0$ 与 k, m 无关， s.t.

$$\|I-B_kA_k\|_A \le \delta := \frac{\alpha}{\alpha+m}$$

$$\|I-B_kA_k\|_A \le \sup_{0\neq v\in V_k} \frac{\|(I-B_kA_k)v\|_A}{\|v\|_A} = \sup_{0\neq v\in V_k} \frac{a((I-B_kA_k)v, v)}{\|v\|_A^2}$$

$$\|u^{n+1}-u_k\|_A \le \delta\, \|u^n-u_k\|_A$$

证明：先证 $0\le a((I-B_kA_k)v, v)\le \delta\, a(v, v)$

假设 $k-1$ 时成立

$$a((I-B_kA_k)v, v) = \underbrace{\|(I-P_{k-1})K_k^m v\|_A^2}_{\ge 0} + a((I-B_{k-1}A_{k-1})P_{k-1}K_k^m v, P_{k-1}K_k^m v)$$

$$\le \delta(\|K_k^m v\|_A^2 - \|(I-P_{k-1})K_k^m v\|_A^2)$$

$$a((I-B_kA_k)v, v) \le \|(I-P_{k-1})K_k^m v\|_A^2 + \underline{\delta\,\|P_{k-1}K_k^m v\|_A^2}$$

由 $\|(I-P_{k-1})K_k^m v\|_A^2 \le \frac{\alpha}{m}(\|v\|_A^2 - \|K_k^m v\|_A^2)$

$$\|v\|_A^2 = \|P_{k-1}v\|_A^2 + \|(I-P_{k-1})v\|_A^2$$

$$\le (1-\delta)\frac{\alpha}{m}(\|v\|_A^2 - \|K_k^m v\|_A^2) + \delta\,\|K_k^m v\|_A^2$$

$$\le \delta\,\|v\|_A^2$$

每步 $O(n_k)$ $\quad \|u^{n+1}-u_k\|_A \le \delta\,\|u^n-u_k\|_A$

$$\|u_k-u\|_A = O(h_k)$$

$$\delta^n\,\|u^0-u_k\|_A \qquad \delta^n \le Ch_k.$$

$$n \ge \frac{\log(Ch_k)}{\log\delta} = O(\log h_k^{-1}) = O(\log n_k)$$

MG法计算量 $O(n_k\log n_k)$

§4 完全 MG（FMG）方法

$$A_ku_k = f_k$$

FMG： $k=1$ 时 $\hat{u}_1 = A_1^{-1}f_1$

$k>1$， 令 $\hat{u}_k = \hat{u}_{k-1}$ 作MG迭代 L 次.

$$\hat{u}_k \leftarrow \hat{u}_k + B_k(f_k - A_k\hat{u}_k)$$

$\exists\, p$, s.t. $C_2p^{-k}\le h_k\le C_3p^{-k}$ $\quad p$是某个常数 衡量加密效果？

$$\|u_k-\hat{u}_k\|_A \le \delta^L\,\|u_k-\hat{u}_{k-1}\|_A$$

$$\|u_k-u_{k-1}\|_A \le \|u-u_{k-1}\|_A \le C_1h_{k-1}$$

$$\le \delta^L(\|u_{k-1}-\hat{u}_{k-1}\|_A + \|u_k-u_{k-1}\|_A)$$

$$\|u-u_k+\underbrace{u_k-u_{k-1}}_{\in V_k}\|_A$$

$$\le \delta^L(\|u_{k-1}-\hat{u}_{k-1}\|_A + C_1h_{k-1})$$

$$\leq \delta^L (\|u_{k-1} - \hat{u}_{k-1}\|_A + C_1C_3 p^{1-k})$$

$$\vdots$$

$$\leq C_1C_3 (\delta^L p^{1-k} + \delta^{2L} p^{2-k} + \dots + \delta^{(k-1)L} p)$$

$$\leq C_1C_3 p^{-k} (\delta^L p + \dots + (\delta^L p)^{k-1})$$

定理：设 $\delta^L p < 1$ 则. $\|u_k - \hat{u}_k\|_A \leq \frac{C_1C_3}{C_2} \frac{\delta^L p}{1-\delta^L p} \cdot h_k \leq C h_k$.

$W_k = O(mn_k) + W_{k-1}$

$W_k = O(n_1 + \dots + n_k)$ $\quad n_k = O(h_k^{-d}) = O(p^{dk})$

$= O(n_k)$

$v_k = v_k^1\phi_k^1 + \dots + v_k^{n_k}\phi_k^{n_k}$

$\tilde{v}_k = (v_k^i)_{n_k \times 1}$ $\quad (\tilde{\tilde{f}}_k)_i = (f_k, \phi_k^i)$

$\widetilde{\widetilde{A_k v_k}} = \tilde{A}_k \tilde{v}_k$

(FEM) $\quad \tilde{A}_k \tilde{u}_k = \tilde{\tilde{f}}_k$ $\quad AU = F$

MG: 给定 $u^0 \in V_k$ $\quad \tilde{u}^{n+1} = \tilde{u}^n + \tilde{B}_k(\tilde{\tilde{f}}_k - \tilde{A}_k\tilde{u}^n)$ $\quad n = 0, 1, 2, \dots$

$u_k = u_k + B_k(f_k - A_k u_k)$

MGP: 定义 B_k：给定 m, R_k：设 $\tilde{B}_1 = \tilde{A}_1^{-1}$

$\tilde{B}_{k-1}: V_{k-1} \longrightarrow V_{k-1}$ 已定义 $\quad \tilde{\tilde{g}} \in R^{n_k}$ 如下定义：$\tilde{B}_k\tilde{g}$

$A_k y = g$ $\quad$ 0为初值 $\Leftrightarrow B_k g$.

(i) 前磨光：$\tilde{y}^0 = 0$ $\quad \tilde{y}^{i+1} = \tilde{y}^i + R_k(\tilde{\tilde{g}} - \widetilde{A_k y^i})$ $\quad i = 0 \dots m-1$

(ii) 粗网格校正： $\quad \tilde{e} = \tilde{B}_{k-1} \cancel{Q_{k-1}} (I_{k-1}^k)^T (\tilde{\tilde{g}} - \widetilde{A_k y^m})$ $\quad \tilde{y}^{m+1} = \tilde{y}^m + \cancel{e}\ I_{k-1}^k\tilde{e}$

(iii) 后磨光： $\quad \tilde{y}^{i+1} = \tilde{y}^i + \tilde{R}_k^t(\tilde{g} - \widetilde{A_k y^i})$ $\quad i = m+1, \dots, 2m$

$(R_k v_k, w_k) = (v_k, R_k^t w_k)$

$\tilde{B}_k\tilde{\tilde{g}} = \tilde{y}^{2m+1}$

定义 $I_{k-1}^k \in R^{n_k \times n_{k-1}}$ $\quad$ Prolongation Matrix

$$\phi_{k-1}^j = \sum_{i=1}^{n_k} (I_{k-1}^k)_{ij} \phi_k^i$$

引理： (i) $v_k \in V_k$. $v_{k-1} \in V_{k-1}$ $\quad v_k = v_{k-1}$. $\quad \tilde{v}_k = I_{k-1}^k \tilde{v}_{k-1}$

(ii) $\forall\ r_k \in V_k$

(iii) $\widehat{A}_{k-1} = (I_{k-1}^k)^T \widehat{A}_k I_{k-1}^k$ $\quad \widetilde{\widetilde{Q_{k-1} r_k}} = (I_{k-1}^k)^T \widetilde{\widetilde{r}}_k$

(ii) $(\widetilde{\widetilde{Q_{k-1} r_k}})_j = (\cancel{Q_{k-1}} r_k \cdot \phi_{k-1}^j) = \sum_{i=1}^{n_k} (I_{k-1}^k)_{ij} (r_k \cdot \phi_k^i)$

$Q_{k-1} A_k = A_{k-1} P_{k-1}$ $\quad = \cdots\ (\widetilde{\widetilde{r}}_k)_i$

(iii) $\underset{\widehat{V}_{k-1}}{v_{k-1}} = \underset{\widehat{V}_k}{v_k}$ $\quad Q_{k-1} A_k v_{k-1} = A_{k-1} P_{k-1} v_{k-1} = A_{k-1} v_{k-1}$

$$\widetilde{\widetilde{Q_{k-1} A_k v_k}} = \widetilde{\widetilde{A_{k-1} v_{k-1}}}$$

$$(I_{k-1}^k)^T \widetilde{\widetilde{A_k v_k}} = \widehat{A}_{k-1} \widetilde{v}_{k-1}$$

$$(I_{k-1}^k)^T \widehat{A}_k \underbrace{\widetilde{v}_k}_{} = \widehat{A}_{k-1} \widetilde{v}_{k-1} \qquad \checkmark$$

$I_{k-1}^k \tilde{v}_{k-1}$

FMG: $k=1$ 时. $\widetilde{\widehat{u}}_1 = \widetilde{A}_1^{-1} \widetilde{f}_1$

$k>1$. 令 $\widetilde{\widehat{u}}_k = I_{k-1}^k \widetilde{\widehat{u}}_{k-1}$ $\quad$ 作MG迭代 ℓ 次.

$$\widetilde{\widehat{u}}_k \leftarrow \widetilde{\widehat{u}}_k + \widetilde{B}_k (\widetilde{\widetilde{f}}_k - \widetilde{A}_k \widetilde{\widehat{u}}_k)$$